

SSIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	Ch.saisri	Reg. No.:	192472367
Section / Batch:	Cse-ai	Date:	

Code of Conduct

I **ch.saisri** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

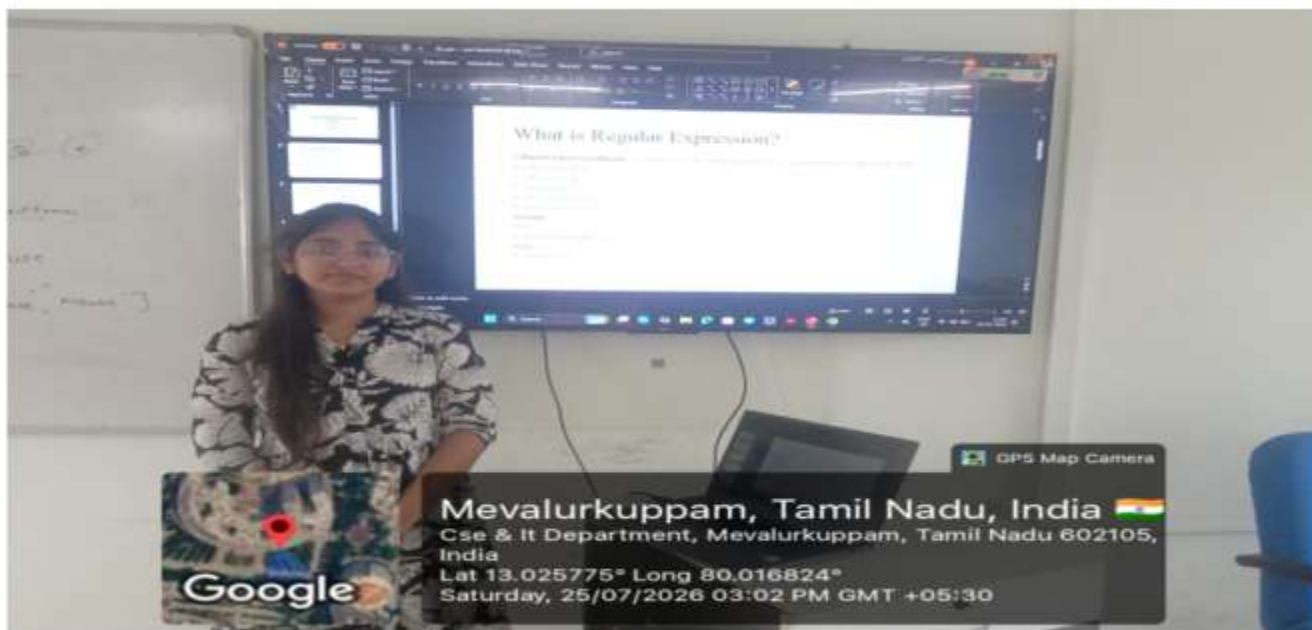
- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Applications of Morphological Parsing in Natural Language Processing

DSAO308- NATURAL LANGUAGE PROCESSING

Presented By CH.SAISRI(192472367)

GUIDED BY KUMARAGURUBARAN. T

Introduction

Definition

Morphological Parsing is the process of analyzing the internal structure of words by dividing them into **morphemes**, the smallest meaningful units of language.

Why is it needed?

Understands word structure.
Identifies root words and affixes.
Extracts grammatical information.
Improves language understanding.

Example

Unhappiness → Un + Happy + ness

Architecture of a Morphological Parser



Description

- Tokenizer:** Splits text into words.
- Lexicon:** Stores root words.
- Rule Engine:** Applies prefix/suffix rules.
- Feature Extractor:** Finds tense, number, gender, etc.
- Output:** Produces the analyzed word.

Morphological Features Extracted

The parser identifies:
Stem (Root Word)
Prefix
Suffix
Part of Speech (POS)
Number (Singular/Plural)
Tense
Person
Gender (*where applicable*)

Word	Extracted Features
Running	Stem = Run, POS = Verb, Tense = Present Continuous
Cats	Stem = Cat, Number = Plural
Unhappy	Prefix = Un, Stem = Happy

Morphological Parsing Techniques

1. Rule-Based Parsing

Uses predefined grammar rules.
Fast and simple.
Best for regular word forms.

2. Dictionary-Based Parsing

Compares words with a lexicon.
Handles irregular words.

3. Finite-State Transducer (FST)

Uses states and transitions.
Efficient for large vocabularies.

4. Machine Learning-Based Parsing

Learns patterns from training data.
More flexible for complex languages.

Applications of Morphological Parsing

Application	Purpose
Information Retrieval	Improves search results
Machine Translation	Produces accurate translations
Spell Checking	Detects and corrects spelling errors
Text Processing	Helps summarize and classify text
Speech Processing	Improves speech recognition

Real-World Applications

Examples

- 🔍 **Google Search** – Finds related word forms.
- 🌐 **Google Translate** – Understands grammar before translation.
- ✍️ **Grammarly** – Checks spelling and grammar.
- 💬 **ChatGPT** – Analyzes word structure to understand user input.
- 🗣️ **Siri & Alexa** – Improve speech recognition

Advantages and Challenges

Advantages

- Improves language understanding.
- Reduces vocabulary size.
- Increases NLP accuracy.
- Supports multilingual systems.

Challenges

- Irregular words.
- Ambiguous meanings.
- Different grammar rules across languages.
- Complex morphology in languages like Tamil or Arabic.

Future Scope

Emerging Areas

- AI-powered chatbots
- Large Language Models (LLMs)
- Multilingual NLP
- Medical document analysis
- Legal document processing
- Intelligent voice assistants

Conclusion

- Morphological parsing analyzes the internal structure of words.
 - It extracts grammatical and semantic information.
 - It improves the performance of many NLP applications.
 - It is a key technology behind modern language-processing systems.
-

Thank you

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood

Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %
1	20	25	20	12	7	84

Faculty In-charge	Course Coordinator	Program Director
Dr.T.Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____